

MehtaOmPset3

April 14, 2025

Problem Set 2

Om Mehta

FINM 37400

1. Hotelling's T^2 - small dataset

a)

$$X = \begin{bmatrix} 2 & 12 \\ 8 & 9 \\ 6 & 9 \\ 8 & 10 \end{bmatrix}, \mu_0 = \begin{bmatrix} 7 \\ 11 \end{bmatrix}$$

```
[47]: import numpy as np
```

```
[48]: X = np.array([
        [2, 12],
        [8, 9],
        [6, 9],
        [9, 10]
    ])

    xbar = np.mean(X, axis=0)
    S = np.cov(X.T)
    S_inv = np.linalg.inv(S)
    xbar, S, S_inv
```

```
[48]: (array([ 6.25, 10.  ]),
      array([[ 9.58333333, -3.33333333],
             [-3.33333333,  2.          ]]),
      array([[0.24827586, 0.4137931 ],
             [0.4137931 , 1.18965517]]))
```

```
[49]: n = 4
      mu_0 = np.array([7, 11]).T
      meandiff = xbar - mu_0
```

```
T2 = n * (meandiff.T @ S_inv @ meandiff)
round(T2, 2)
```

[49]: 7.8

$$T^2 \sim \frac{(n-1)p}{n-p} F_{p, n-p}(\alpha)$$

$$T^2 \sim 3 \cdot F_{2,2}$$

b)

```
[50]: C = np.array([
        [1, 1],
        [-1, 1]
    ])

Y = X @ C.T
y_bar = np.mean(Y, axis=0)
S_y = C @ S @ C.T
mu_0_star = C @ mu_0
diff = y_bar - mu_0_star
T2 = n * (diff.T @ np.linalg.inv(S_y) @ diff)
round(T2, 2)
```

[50]: 7.8

c)

$$T_y^2 = n(\bar{y} - \mu_y)^\top S_y^{-1}(\bar{y} - \mu_y)$$

With:

$$\bar{y} = C\bar{x}$$

$$\mu_y = C\mu_0$$

$$S_y = CSC^\top$$

$$T_y^2 = n(C(\bar{x} - \mu_0))^\top (CSC^\top)^{-1} (C(\bar{x} - \mu_0))$$

Since C is invertible, $(CSC^\top)^{-1} = (C^\top)^{-1}S^{-1}C^{-1}$

$$T_y^2 = n(C(\bar{x} - \mu_0))^\top (C^\top)^{-1}S^{-1}C^{-1}(C(\bar{x} - \mu_0))$$

Let

$$d = \bar{x} - \mu_0$$

$$\Rightarrow T_y^2 = n(Cd)^\top (C^\top)^{-1}S^{-1}C^{-1}(Cd)$$

$$\text{Recall: } (Cd)^\top = d^\top C^\top$$

$$\text{So: } T_y^2 = n \cdot d^\top C^\top (C^\top)^{-1}S^{-1}C^{-1}Cd$$

$$\text{Substitute: } C^\top (C^\top)^{-1} = I, \quad C^{-1}C = I$$

$$\text{So it collapses to: } T_y^2 = n(\bar{x} - \mu_0)^\top S^{-1}(\bar{x} - \mu_0)$$

2. Derivation in Canonical Correlation Analysis

$$A = \Sigma_{11}^{-1}\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

$$\Sigma_{11}^{-1} = \frac{1}{1-p^2} \begin{bmatrix} 1 & -p \\ -p & 1 \end{bmatrix}$$

$$\Sigma_{22}^{-1} = \frac{1}{1-q^2} \begin{bmatrix} 1 & -q \\ -q & 1 \end{bmatrix}$$

$$\Sigma_{12} = \Sigma_{21} = \begin{bmatrix} r & r \\ r & r \end{bmatrix}$$

$$M = \Sigma_{11}^{-1} \Sigma_{12} = \frac{r-pr}{1-p^2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$M \cdot \Sigma_{22}^{-1} = \frac{r-pr}{1-p^2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -q \\ -q & 1 \end{bmatrix}$$

$$= \frac{r-pr}{1-p^2} \cdot \frac{1-q}{1-q^2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$A = \frac{r-pr}{1-p^2} \cdot \frac{1-q}{1-q^2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} r & r \\ r & r \end{bmatrix}$$

$$= \frac{r-pr}{1-p^2} \cdot \frac{1-q}{1-q^2} \cdot 2r \cdot \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$= \frac{2r^2(1-p)(1-q)}{(1-p^2)(1-q^2)} \cdot \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\lambda_1 = 2 \cdot \frac{2r^2(1-p)(1-q)}{(1-p^2)(1-q^2)}$$

$$\lambda_2 = 0$$

$$\rho_1^* = \sqrt{\lambda_1} = \sqrt{\frac{2r^2(1-p)(1-q)}{(1-p^2)(1-q^2)}}$$

b)

$$a_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$a_1^\top \Sigma_{11} a_1 = 1$$

$$\Rightarrow \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & p \\ p & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= 1 + p + p + 1 = 2(1+p)$$

$$\Rightarrow a_1 = \frac{1}{\sqrt{2(1+p)}} \begin{bmatrix} 1 & p \\ p & 1 \end{bmatrix}$$

$$b_1 = \Sigma_{22}^{-1} \Sigma_{21} a_1$$

$$\Sigma_{21} a_1 = \frac{2r}{\sqrt{2(1+p)}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{aligned} b_1 &= \frac{2r}{\sqrt{2(1+p)}} \cdot \frac{1}{1-q^2} \begin{bmatrix} 1 & -q \\ -q & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ &= \frac{2r(1-q)}{(1-q^2)\sqrt{2(1+p)}} \end{aligned}$$

$$\Rightarrow b_1^\top \Sigma_{22} b_1 = 1$$

$$\Rightarrow b_1 = \frac{1}{\sqrt{2(1+q)}} \begin{bmatrix} 1 & q \\ q & 1 \end{bmatrix}$$

$$a_1 = \frac{1}{\sqrt{2(1+p)}} \begin{bmatrix} 1 & p \\ p & 1 \end{bmatrix}, \quad b_1 = \frac{1}{\sqrt{2(1+q)}} \begin{bmatrix} 1 & q \\ q & 1 \end{bmatrix}$$

$$\Rightarrow (U_1, V_1) = (a_1^\top X, b_1^\top Y)$$

3. Practice CCA, Low Dimensions

```
[78]: import pandas as pd
      from sklearn.cross_decomposition import CCA
      from matplotlib import pyplot as plt
```

```
[79]: df = pd.read_csv("stiffness.dat", sep=r"\s+", header=None)
      df.columns=['x1', 'x2', 'x3', 'x4', 'd2']
```

```
[80]: X, Y, d2 = df[['x1', 'x2']], df[['x3', 'x4']], df['d2']
```

a)

```
[81]: cca = CCA(n_components=2)
      X_c, Y_c = cca.fit_transform(X, Y)
```

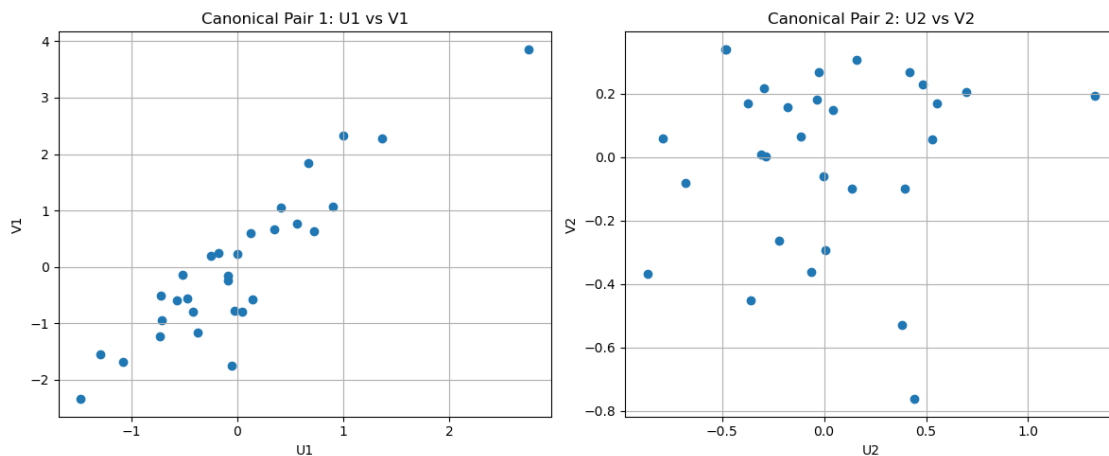
b)

```
[82]: U, V = cca.x_weights_, cca.y_weights_  
      U[0], V[0]
```

```
[82]: (array([0.98710551, 0.16007097]), array([0.55489103, 0.83192304]))
```

c)

```
[83]: fig, ax = plt.subplots(1, 2, figsize=(12, 5))  
      ax[0].scatter(X_c[:, 0], Y_c[:, 0])  
      ax[0].set_title("Canonical Pair 1: U1 vs V1")  
      ax[0].set_xlabel("U1")  
      ax[0].set_ylabel("V1")  
      ax[0].grid(True)  
  
      ax[1].scatter(X_c[:, 1], Y_c[:, 1])  
      ax[1].set_title("Canonical Pair 2: U2 vs V2")  
      ax[1].set_xlabel("U2")  
      ax[1].set_ylabel("V2")  
      ax[1].grid(True)  
  
      plt.tight_layout()  
      plt.show()
```



d)

```
[84]: rho1 = np.corrcoef(X_c[:, 0], Y_c[:, 0])[0, 1]  
      rho2 = np.corrcoef(X_c[:, 1], Y_c[:, 1])[0, 1]  
      print('Canonical correlation 1:', round(rho1, 2))  
      print('Canonical correlation 2:', round(rho2, 2))
```

Canonical correlation 1: 0.91
Canonical correlation 2: 0.07

Based on these correlations, it seems that the **first canonical pair** holds the important correlation. There is a strong linear relationship between a certain linear combination of the vibrational and static tests, suggesting that one aspect of stiffness captured by the dynamic methods is highly predictive of a similar aspect measured through static testing.

4. Multivariate vs. Univariate Inference

```
[85]: df = pd.read_csv("fly.dat", sep=r'\s+')  
df
```

```
[85]:
```

	Species	Ant.Length	Wing.Length
1	Af	1.38	1.64
2	Af	1.40	1.70
3	Af	1.24	1.72
4	Af	1.36	1.74
5	Af	1.38	1.82
6	Af	1.48	1.82
7	Af	1.54	1.82
8	Af	1.38	1.90
9	Af	1.56	2.08
10	Apf	1.14	1.78
11	Apf	1.20	1.86
12	Apf	1.18	1.96
13	Apf	1.30	1.96
14	Apf	1.26	2.00
15	Apf	1.28	2.00

```
[86]: df['Y1'] = df['Ant.Length'] + df['Wing.Length']  
df['Y2'] = df['Wing.Length']  
df_af = df[df['Species'] == 'Af']  
df_apf = df[df['Species'] == 'Apf']  
Y_af = df_af[['Y1', 'Y2']].to_numpy()  
Y_apf = df_apf[['Y1', 'Y2']].to_numpy()
```

```
[87]: ybar_af, ybar_apf = Y_af.mean(axis=0), Y_apf.mean(axis=0)  
meandiff = ybar_af - ybar_apf  
S_af = np.cov(Y_af.T)  
S_apf = np.cov(Y_apf.T)  
n1, n2 = len(Y_af), len(Y_apf)  
S_p = ((n1-1)*S_af + (n2-1)*S_apf) / (n1+n2-2)  
T2 = ((n1*n2)/(n1+n2)) * (meandiff.T @ np.linalg.inv(S_p) @ meandiff)
```

```
[88]: from scipy.stats import f
```

```
[93]: p = 2
      F = ((n1 + n2 - p - 1) / p * (n1 + n2 - 2)) * T2
      df1, df2 = 2, n1 + n2 - p - 1
      pvalue = 1 - f.cdf(F, df1, df2)
```

```
[94]: pvalue
```

```
[94]: 8.260059303211165e-14
```

The low p-value indicates that we can reject the null hypothesis of equal mean vectors, concluding that there is a **statistically significant difference in multivariate means of Y_1 and Y_2 between the AF and APF species.**

Part ii. Yes, we can expect the same result from X_1 and X_2 because the transformation to Y_1 and Y_2 is **linear and invertible.**

b)

```
[95]: from scipy.stats import ttest_ind

      t_stat_y1, p_y1 = ttest_ind(df_af['Y1'], df_apf['Y1'], equal_var=True)
      t_stat_y2, p_y2 = ttest_ind(df_af['Y2'], df_apf['Y2'], equal_var=True)
      print("p-value for Y1:", round(p_y1, 4))
      print("p-value for Y2:", round(p_y2, 4))
```

```
p-value for Y1: 0.5202
```

```
p-value for Y2: 0.0663
```

Though neither of the *univariate* tests for Y_1 and Y_2 yielded significance at the $\alpha = 0.01, 0.05$ levels, Hotelling's T^2 test from part a) detected a statistically significant difference in the joint distribution of Y_1 and Y_2 . This indicates that **multivariate tests reveal more structure that univariate tests miss, especially when variables are correlated.**

c)

```
[96]: plt.figure(figsize=(8, 6))

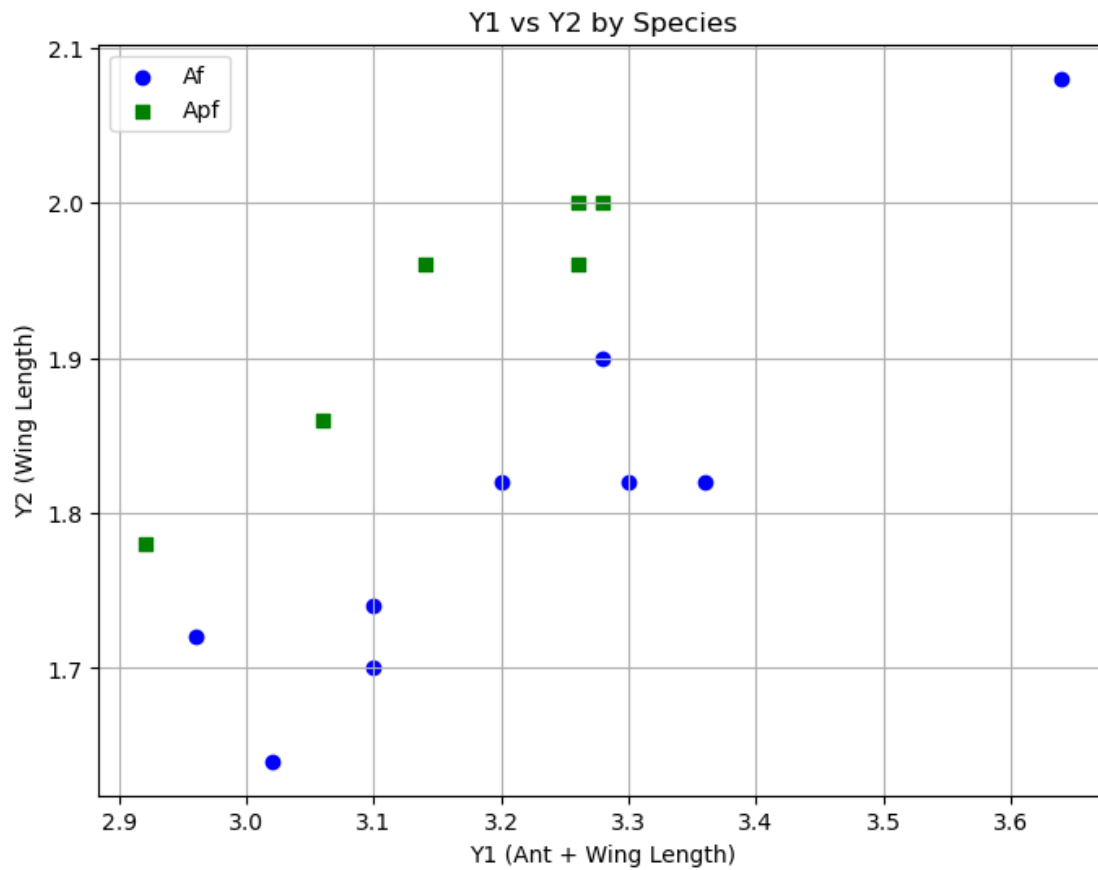
      # Species Af
      plt.scatter(df_af['Y1'], df_af['Y2'], color='blue', marker='o', label='Af')

      # Species Apf
      plt.scatter(df_apf['Y1'], df_apf['Y2'], color='green', marker='s', label='Apf')

      # Labels and title
      plt.xlabel("Y1 (Ant + Wing Length)")
      plt.ylabel("Y2 (Wing Length)")
      plt.title("Y1 vs Y2 by Species")
      plt.grid(True)
      plt.legend()
```



```
plt.show()
```



d)

```
[104]: fig, ax = plt.subplots(figsize=(8, 6))

ax.plot(mean_diff[0], mean_diff[1], 'ko', label='Mean Difference')

ellipse = Ellipse(xy=mean_diff, width=width, height=height,
                  angle=angle, edgecolor='red', fc='None', lw=2, label='98% CI')
    ↳Ellipse')
ax.add_patch(ellipse)

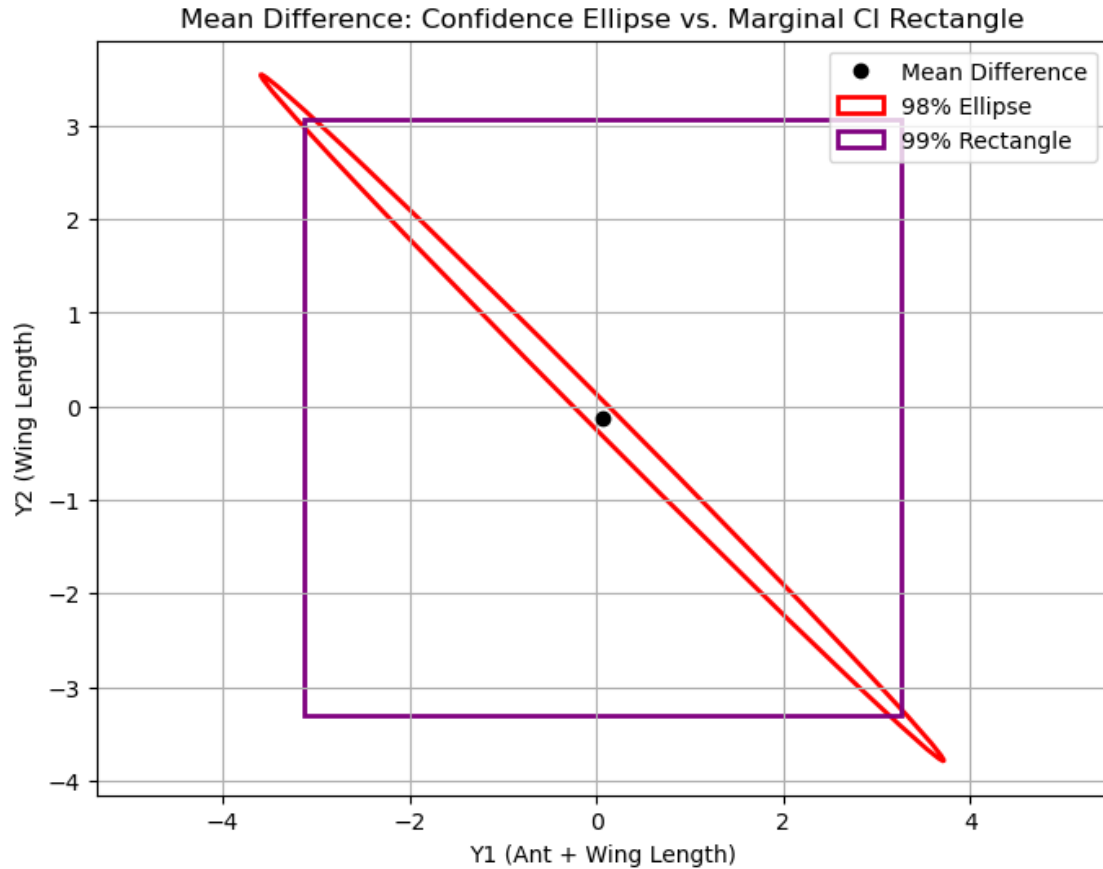
rectangle = Rectangle(lower_left, rect_width, rect_height,
                     edgecolor='purple', fc='None', lw=2, label='99% CI')
    ↳Rectangle')
ax.add_patch(rectangle)

ax.set_xlabel("Y1 (Ant + Wing Length)")
```

```

ax.set_ylabel("Y2 (Wing Length)")
ax.set_title("Mean Difference: Confidence Ellipse vs. Marginal CI Rectangle")
ax.grid(True)
ax.legend()
plt.axis('equal')
plt.show()

```



iii. The rectangle in part ii.) is constructed from two separate 99% marginal confidence intervals—one for the mean difference in Y_1 , and one for the mean difference in Y_2 . Using the Bonferroni correction for two comparisons, we adjust the per-variable confidence level to ensure that the overall (joint) confidence level is at least 98%.

iv. The zero vector is inside both the ellipse and the rectangle, implying that at both levels we fail to reject the null hypothesis that the true species mean difference is zero.

5. Vector-matrix derivations for multivariate data and random variables

a)

$$\text{Let: } X_c = X - \mathbf{1}_n \bar{x}^\top$$

$$\bar{x} = \frac{1}{n} \mathbf{1}_n^\top X \Rightarrow X_c = X - \frac{1}{n} \mathbf{1}_n \mathbf{1}_n^\top X$$

$$X_c = HX \quad \text{where} \quad H = I_n - \frac{1}{n} \mathbf{1}_n \mathbf{1}_n^\top$$

$$S = \frac{1}{n-1} X_c' X_c = \frac{1}{n-1} (HX)' (HX) = \frac{1}{n-1} X' H' H X$$

$$H' = H \quad \text{and} \quad H^2 = H \quad \Rightarrow \quad S = \frac{1}{n-1} X' H X$$

b)

$$\text{Cov}[W] = \mathbb{E}[(W - \mathbb{E}[W])(W - \mathbb{E}[W])^\top]$$

Substitute $W = AY + c$, so $\mathbb{E}[W] = A\mathbb{E}[Y] + c$

$$\Rightarrow \text{Cov}[W] = \mathbb{E}[(A(Y - \mathbb{E}[Y]))(A(Y - \mathbb{E}[Y]))^\top]$$

$$= \mathbb{E}[A(Y - \mathbb{E}[Y])(Y - \mathbb{E}[Y])^\top A^\top]$$

$$= A \mathbb{E}[(Y - \mathbb{E}[Y])(Y - \mathbb{E}[Y])^\top] A^\top$$

$$= A \text{Cov}[Y] A^\top$$

c)

$$\text{Cov}(\mathbf{a}^\top \mathbf{Y}, \mathbf{b}^\top \mathbf{W}) = \mathbb{E}[(\mathbf{a}^\top \mathbf{Y} - \mathbb{E}[\mathbf{a}^\top \mathbf{Y}])(\mathbf{b}^\top \mathbf{W} - \mathbb{E}[\mathbf{b}^\top \mathbf{W}])]$$

$$= \mathbb{E}[\mathbf{a}^\top (\mathbf{Y} - \mathbb{E}[\mathbf{Y}]) \cdot (\mathbf{W} - \mathbb{E}[\mathbf{W}])^\top \mathbf{b}]$$

$$= \mathbf{a}^\top \mathbb{E}[(\mathbf{Y} - \mathbb{E}[\mathbf{Y}])(\mathbf{W} - \mathbb{E}[\mathbf{W}])^\top] \mathbf{b}$$

$$= \mathbf{a}^\top \text{Cov}(\mathbf{Y}, \mathbf{W}) \mathbf{b}$$